THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sum of the first 9 odd numbers is: 1

(a) 64 **(b)** 90 **(c)** 72 **(d)** 81

2. A complete graph with 5 points has how many lines? 1

(a) 5 **(b)** 20 **(c)** 15 **(d)** 10

3. The angle between the hands of a clock at 6 o'clock is: 1

(a) 360° **(b)** 180° **(c)** 90° **(d)** 120°

4. An angle of 181° is: 1

(a) right **(b)** reflex **(c)** acute **(d)** obtuse

5. The Collatz sequence starting at 20 takes how many steps to reach 1? 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 10

6. The number of days between 5 Jan and 20 Jan is: 1

(a) 15 **(b)** 20 **(c)** 16 **(d)** 14

7. In a bar graph with bars 20, 35, 15 and 30, the tallest bar is: 1

(a) 20 **(b)** 15 **(c)** 35 **(d)** 30

8. Each separate piece of information in data is called an: 1

(a) observation **(b)** angle **(c)** axis **(d)** edge

9. The HCF of 9 and 12 is: 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 2

10. Which of these is a composite number? 1

(a) 47 **(b)** 53 **(c)** 51 **(d)** 59

11. Area of a triangle with base 16 cm and height 7 cm is: 1

(a) 23 cm² **(b)** 112 cm² **(c)** 56 cm² **(d)** 72 cm²

12. The cost of painting a floor depends on its: 1

(a) perimeter **(b)** angle

(c) number of corners **(d)** area

13. Which is greater: $\frac{5}{6}$ or $\frac{7}{9}$? 1

(a) $\frac{5}{6}$ **(b)** cannot be compared

(c) both are equal **(d)** $\frac{7}{9}$

14. On the number line, $\frac{1}{2}$ lies: 1

(a) midway between 0 and 1 **(b)** at 1
(c) at 0 **(d)** beyond 1

15. A rectangle of sides 10 cm and 6 cm has perimeter: 1

(a) 60 cm **(b)** 32 cm **(c)** 34 cm **(d)** 16 cm

16. The set square has angles: 1

(a) 100°, 40°, 40° **(b)** 90°, 45°, 45° or 90°, 30°, 60°
(c) 60°, 60°, 120° **(d)** 90°, 90°, 90°

17. An isosceles triangle has: 1

(a) 2 lines **(b)** 0 lines
(c) 1 line of symmetry **(d)** 3 lines

18. How many lines of symmetry does a rectangle (not a square) have? 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 4

19. The value of $12 + (-15)$ is: 1

(a) −5 **(b)** −3 **(c)** 3 **(d)** −1

20. Numbers less than zero are called: 1

(a) positive integers **(b)** negative integers
(c) fractions **(d)** prime numbers

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Why are the numbers 1, 4, 9, 16, … called square numbers? 2

22. What is a palindrome? Give two examples. 2

23. Find the range of 14, 9, 22, 17, 30. 2

24. List the factors of 24. 2

25. Find the perimeter of an equilateral triangle of side 13 cm. 2

26. Write three fractions equivalent to $\frac{2}{5}$. 2

27. What is the radius of a circle? 2

28. What does a red token and a yellow token together represent? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Why do mathematicians study patterns? 3

30. What is a complete angle? How many right angles does it make? 3

31. Out of 50 students, 18 like cricket, 12 football, 8 hockey and the rest badminton. Find the number liking badminton. 3

32. A rectangular field is 100 m by 60 m. Find the length of its boundary and its area. 3

33. How many different rectangles of whole-number sides can be made with perimeter 16 cm? List them. 3

34. What is meant by ‘order of rotational symmetry’? Give one example. 3

35. Show that 143 is composite by finding its factors. 3

OR

What is the smallest number that leaves remainder 1 when divided by 2, 3 and 4? 3

36. Explain with an example why $(-a) + a = 0$. 3

OR

A submarine at −200 m rises 75 m, then dives 40 m. Where is it? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) State the sum of angles at a point. 5
(ii) If three angles at a point are 100°, 120° and x, find x.
(iii) Name the type of x.

OR

(i) Describe an experiment using paper folding to get a right angle. 5
(ii) Use it to find an angle of 45°.

38. (i) Write the rule for digit sum. 5
(ii) Find the digit sum of 12, 345.
(iii) Test divisibility by 3 for 12345.

OR

(i) How is the number line used for comparing numbers? 5
(ii) Order 4,508; 4,058; 4,805; 4,850.

39. (i) Write $\frac{3}{4}$ of a day in hours. 5
(ii) Write $\frac{2}{3}$ of an hour in minutes.
(iii) What is $\frac{5}{8}$ of 1 kg in grams?

OR

(i) Why are $\frac{2}{4}$ and $\frac{1}{2}$ equal? 5
(ii) Show with folding paper.
(iii) Find the fraction equal to $\frac{1}{2}$ with denominator 18.

40. (i) State three regular polygons with their orders of rotational symmetry. 5
(ii) Find the sum of angles of rotation for one full turn in a pentagon.

OR

(i) Complete the figure given its left half and a vertical mirror line. 5
(ii) State the symmetry.
(iii) Name a real object like it.

END OF QUESTION PAPER — Total: 80 Marks

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